Chinmay Joshi

Machine Learning (Data Science, Efficiency and Security)

chinmay.joshi23@outlook.com · +49 178 8266100 · Saarbrücken · chinmayj23.github.io · linkedin.com/in/chinmay-joshi23

Education

M.Sc. Computer Science — Grade 1.7 (German scale, 1.0 highest) Oct 2022 – May 2026
Universität des Saarlandes *Saarbrücken, Germany*

- Thesis: *On the Effect of Data Pruning on Memorization.* Advisor: Dr. Rebekka Burkholz, CISPA Helmholtz Center for Information Security.
- Recipient of the SaarlandStipendium and the Santander Scholarship.

B.Tech. Data Science — GPA 3.8/4.0 2018 – 2022
Narsee Monjee Institute of Management Studies *Mumbai, India*

Publications

- **Chinmay Joshi**, Advait Gadhikar, Celia Rubio-Madrigal, Aneet Kumar Dutta, Mridula Singh, Rebekka Burkholz. "Prune to Protect: Faster Training and Enhanced Privacy by Dynamic Data Pruning." *ICML 2026 MemFM Workshop; under review, NeurIPS 2026.* [Paper]
- **Chinmay Joshi**, David Antony Selby, Ayush Patnaik, Sebastian Vollmer, Gerrit Großmann. "Dolphin: Interpretable Discovery of Exceptional Subgroups in Longitudinal Data." *NFMCP 2026 Workshop at ECML-PKDD.* [Code]
- **Chinmay Joshi**, Siba Panda. "PCA-LSTM: Deep Learning Approach for the Indian Large-Caps." *IEEE Int. Conference for Convergence in Technology (I2CT), 2022.* [Paper] [Code]

Experience

Research Assistant Dec 2025 – Present
German Research Center for Artificial Intelligence (DFKI) *Kaiserslautern, Germany*
Subgroup discovery on longitudinal data: finding which subpopulations behave differently from the rest over time, and describing each in human-understandable rules.

- Extending subgroup discovery to longitudinal targets. Standard methods compare a single value per entity; here a subgroup is exceptional through the shape of its trajectory over time — e.g. a steeper climb, a delayed recovery, a flat path inside a growing population — and is returned as an interpretable Boolean rule over the covariates.
- Developing the temporal feature-engineering and evaluation pipeline over large longitudinal economic panels: World Bank country indicators and CMIE monthly household income (150,000 households, 900,000+ observations).

Master's Thesis Researcher May 2025 – May 2026
CISPA Helmholtz Center for Information Security (adv. Dr. Rebekka Burkholz) *Saarbrücken, Germany*
AI models memorise individual training examples, exposing them to privacy attacks. This work asked whether a method adopted to improve model training efficiency can also reduce the privacy risk.

- Demonstrated that dynamic data pruning — dropping examples the model has already learned, in order to train faster — also reshapes the memorisation distribution. The rare, hard, long-tail examples that carry the most privacy risk are memorised less, despite being exposed more often under dynamic data pruning.
- Attributed the late rise in memorisation to the last phase of training, where training returns to the full dataset. Proposed a new method (WLIB), which reweights each example's loss in this phase to preserve privacy.
- Quantified the gain under RMIA membership-inference auditing, which detects whether a given image was in the training set. Attack success (TPR@1%FPR) falls from 13.8% to 4.6% on CIFAR-10 (ResNet-20) and from 19.4% to 17.3% on ImageNet (ViT), at unchanged accuracy and 35% faster training.
- Measured memorisation per example, which requires training many models: 400 per method. A GPU-resident data loader at 1.5 s/epoch versus 4 s/epoch with the standard PyTorch DataLoader on one NVIDIA A6000 makes it tractable.
- Extending the framework to large language models.

Research Assistant Mar 2025 – Sep 2025
Fraunhofer IZFP — URBANIST project *Saarbrücken, Germany*
An app that scores a route by the environmental exposure it subjects the traveller to, using sensors placed around the city.

- Turned raw readings from fixed and mobile city sensors into the environmental measures the app scores routes with: UV index, noise level, particulate concentration and ambient light.
- Built the geospatial layer, interpolating scattered sensor readings into a continuous spatial estimate so that every GPS coordinate along a route carries a value for each measure (OpenStreetMap, GeoPandas, Folium).

Research Intern — Apr 2024 – Aug 2024
National Institute of Informatics (adv. Dr. Satoshi Ikehata) — *Tokyo, Japan*

Novel-view synthesis and relighting from a single image: turning one image of an object into a 3D version that can be viewed from new angles and lit differently.

- Implemented a synthetic dataset-generation pipeline in Blender over Objaverse 3D models. It produced a labelled dataset of 150,000 images at 12 viewpoints and 5 lighting environments per object, each carrying matching depth and surface-normal maps.
- Fine-tuned Zero-1-to-3, an image-to-3D diffusion model, to change viewpoint and lighting together from one input image. Analysed where it fails: shape detail absorbed into surface appearance, and the original lighting persisting after relighting.

Research Assistant — Aug 2023 – Mar 2024
Fraunhofer IZFP — SmartPigHome project — *Saarbrücken, Germany*

An AI system that listens to and watches a pig barn and triggers a distraction game when the animals show early signs of stress.

- Developed an audio annotation tool the veterinarians use to label pig sounds. It replaced a paper-based workflow, cut their labelling effort by roughly 80%, and produced the dataset the later health-prediction work depended on.
- Prototyped object detection and multi-object tracking over barn-camera video (YOLO, DINO) to follow individual animals.

Research Assistant — Jun 2023 – Mar 2024
German Research Center for Artificial Intelligence (DFKI) — *Saarbrücken, Germany*

An effort replacing fixed rural bus lines with shuttles that passengers book on demand, run by a Saarland municipality.

- Developed the scheduling service in Python that decides which shuttle takes which booking and in what order it stops, minimising passenger waiting time within a fixed fleet and seat capacity. Validated in large-scale SUMO traffic simulation across demand levels.

Computer Vision Intern — Jan 2022 – Apr 2022
Aurify Systems Pvt. Ltd. — *Mumbai, India*

Computer vision for shops and warehouses, running on the hardware clients already had on site.

- Delivered the vision stack: product recognition, empty-shelf detection and CCTV activity analysis, each tuned to run on site, alongside the dashboards staff used day to day.
- Used pose estimation to identify suspicious behaviour, reaching 91% accuracy in field tests for trespassing.

Technical Skills

AI efficiency	Data selection and pruning, network pruning, quantisation, GPU-resident data loading.
AI privacy	Membership-inference auditing (RMIA, LiRA), memorisation estimation (Feldman–Zhang).
Computer vision	CNNs, Vision Transformers, object detection (YOLO, DINO), multi-object tracking, pose estimation (AlphaPose 2D, SMPL 3D), 3D reconstruction, novel-view synthesis and relighting (Zero-1-to-3), vision–language models.
Generative AI	Large language models (RAG, pre-training, fine-tuning: LoRA, QLoRA), latent diffusion models, VAEs, synthetic dataset generation.
Data Science	Exploratory analysis, temporal feature engineering, predictive modelling, longitudinal and panel data, statistical testing, subgroup discovery, rule and pattern mining, sensor-stream processing, geospatial analysis, visualisation, A/B testing, causal inference.
Programming	Python, C++, SQL, R.
Frameworks	PyTorch, Hugging Face (Transformers, Diffusers), scikit-learn, OpenCV, pandas, GeoPandas, Blender, SUMO.
Tooling	SLURM, multi-GPU distributed training, Docker, Git, CI/CD, AWS (S3, EC2), Weights & Biases; AI-assisted development (Claude Code, Cursor).
Languages	English (fluent), German (A2, progressing toward B1), Hindi, Marathi.

Projects

- **Hierarchical VAE for image compression.** Latent hierarchy added to a VAE codec, giving $\sim$5 dB PSNR over JPEG at matched bitrate. [Report]
- **Bird call recognition.** MFCC features into a CNN over spectrograms; 82% test accuracy on noisy field recordings.

Teaching and Service

- **Teaching volunteer, Satsang Foundation, Mumbai.** Mathematics, physics and English for students from the tribal community of Aarey Colony.
- **COVID-19 relief volunteer, Satsang Foundation.** Food distribution to households affected by the pandemic.